Chizorom Anyalechi

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Education

University of Guelph Sept 2024 – April 2029 (Expected)
Bachelor of Computer Engineering (Co-op) *Guelph, ON*

- Recipient of $2000 Entrance Scholarship based on a 94% admission average, involvement, and volunteering.
- **Relevant Courses:** Micro Computer Interfacing, Computer Architecture, Digital Systems Design, Data Structures, Object-Oriented Programming, Electrical Circuits

Project Experience

Raspberry Pi Embedded Systems | *C, ARM Assembly, Embedded Linux, GPIO* Sept 2026 – Present
Embedded Systems Laboratory *Guelph, ON*

- Developing low-level **C and ARM assembly software** on Raspberry Pi to control hardware through **GPIO, memory-mapped peripheral registers, and Linux system interfaces**.
- Using **embedded Linux and libgpiod** to configure GPIO devices, execute hardware tests, and troubleshoot low-level **hardware/software interactions**.
- Tracing behaviour across **C code, Linux system interfaces, memory-mapped I/O, and physical hardware** to isolate software and configuration issues during integration.
- Validating expected operation through **structured testing** of software, operating-system interfaces, pin configurations, and peripherals.

AI-Assisted IT Help Desk System | *Python, Flask, SQL, SQLite, Gemini API* September 2026

- Developed a **full-stack ticketing application** in **Python, Flask, and SQLite**, letting users submit, track, filter, and resolve support requests through a persistent workflow.
- Integrated a **Large Language Model (Gemini API)**, designing **prompts** to automatically categorize, prioritize, and recommend troubleshooting steps for each ticket.
- Designed a **SQL data model** for ticket status, category, and priority, supporting efficient filtering and **reporting**.
- Implemented **input validation and failure handling**, testing edge cases such as invalid input and service outages so core features stayed reliable when the AI was unavailable.
- **Coded alongside AI coding assistants** to draft, debug, and refine Python and SQL, testing every change.

16-Bit CPU Design and Implementation | *VHDL, FPGA, Computer Architecture* Jan – Mar 2026

- Designed and implemented a **16-bit CPU** in **VHDL** executing **11 instruction types**, with a full datapath of **ALU, register file, program counter, and memory modules**.
- Built a custom **control unit** for instruction decoding, defining and debugging **9 control signals** for **R-, I-, and J-type instructions**.
- Implemented **branch and jump functionality** by designing **BNE and JMP** target calculation modules for conditional and unconditional control flow.
- Verified the design with **simulation waveforms and edge-case tests**, identifying and resolving control-signal and datapath errors to improve reliability.

Academic and Community Involvement

Formula SAE Racing Team – Electrical Subteam Jan 2025 – Sep 2026
University of Guelph *Guelph, ON*

- Designed, built, and installed the **wiring harness** for a **Formula-style race car**, routing **low-voltage connections** between sensors, controllers, and vehicle systems from **schematics and wiring diagrams**.
- Ensured **electrical reliability and safety** through **continuity testing, troubleshooting, and wire management** in a cross-functional team for **international SAE competitions**.

Volunteer and Work Experience

Summer Camp Activity Coordinator Summer 2023
Heartland Church Connected *Mississauga, ON*

- Supervised over **50 children** daily alongside a staff team, **communicating clearly** with staff and families.
- Planned weekly schedules, coordinated activities, and managed supplies, keeping program operations **organized** and running smoothly.

Technical Skills

- **Programming:** C, C++, ARM Assembly, Python, Java, SQL, VHDL, JavaScript, HTML, CSS, MATLAB
- **Embedded Systems & Hardware:** Raspberry Pi, Arduino, embedded Linux, GPIO, FPGA development, digital logic design, computer architecture, circuit analysis
- **Software & AI:** Flask, SQLite, Pandas, LLM integration (Gemini API), prompt design, PyTorch, AI coding assistants
- **Tools:** Git/GitHub, VS Code, Xilinx ISE, Arduino IDE, libgpiod, AutoCAD, Microsoft Office
- **Core Skills:** Debugging, testing and validation, documentation, problem-solving, communication, teamwork